

Purchase Description

Navy-EXWC

Trailer Lowbed Hydraulic Removable Gooseneck 60 Ton

NOTICE TO OFFERORS:

Offeror shall explicitly and completely identify any and all exceptions taken to the requirements of this specification. Drawings, literature, and any other information submitted with an offer do not constitute a stated or implied exception unless specifically identified in writing as an exception and accepted and implemented by an amendment to the solicitation or modification to a contract. Any exception deemed acceptable by the Government shall only apply to the specific item, requirement, etc. cited, and shall not extend to any other requirement.

Offeror offering other than brand name items identified below, when an “or equal” may be acceptable, shall furnish with their offers all technical data information, product descriptions, etc. to ensure that a determination may be made as to the equality of the product(s) offered (see clause titled Brand “Name or Equal” set forth in Section 52.211-6 of the Federal Acquisition Regulations.)

The following shall be provided with the offer.

1. Detailed explanation, listed in a separate letter, of any exceptions to the specification, including alternate equipment proposed or a declaration of comprehensive understanding of the requirements with no exceptions taken.
2. The offeror shall submit with the offer, preliminary drawings that show the following:
 - a. Layout views of the proposed trailer including all equipment locations and basic dimensions verifying compliance with **ALL** the requirements herein.
 - b. Exterior front, rear, and side view with overall dimensions.
 - c. Estimated curb weight and Gross Trailer Weight Rating (GVWR).
 - d. Documented evidence of quality assurance procedures or system.
 - e. Offeror shall provide access to customers and warranty service, as well as technical support via email, phone service, internet/web, or other method. Offeror shall provide detailed information describing their customer and warranty service access.

1.0 SCOPE and BACKGROUND

Naval Facilities Engineering and Expeditionary Warfare Center (NAVFAC EXWC) has a requirement to replace an aging fleet of trailers based on current mission and support requirements. NAVFAC EXWC has documented requirements to address shortcomings of the current fleet that at a minimum comply with Military (MIL), American

Welding Society (AWS), and Society of American Engineer (SAE) standards for tactical and commercial trailers. Compliance with these government and non-government standards introduces commonality with other military, Department of Defense (DOD), and civilian trailers, allowing fleet and joint operations to utilize the capabilities to their fullest. The trailers will also support operational readiness and functionality in austere conditions.

2.0 APPLICABLE DOCUMENTS

2.1 FEDERAL STANDARDS AND CODES

Code of Federal Regulations 49 CFR 393 (FMCSR) and 571 (FMVSS).

Application for copies of DOT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. DOT standards may be accessed on the internet at <http://www.access.gpo.gov>.

MIL-STD- 209K, Interface Standard for Lift and Tie Down Provisions.

MIL-STD-1366E, Interface Standard for Transportability Criteria.

MIL-STD-1791D_CHG1, Department of Defense Interface Standard for Internal Aerial Delivery in Fixed Wing Aircraft

A-A-59295 (NOTICE 1) COMMERCIAL ITEM DESCRIPTION: CORROSION PREVENTIVE COMPOUNDS, COLD APPLICATION (FOR NEW AND FIELDIED MOTOR VEHICLES AND TRAILERS) (17 NOV 2008) [SUPERSEDING MIL-PRF-62218B]., A-A-59295, dated 09 September 1998

2.2 SOCIETY OF AUTOMOTIVE ENGINEERS, INC

J163 - Low Tension Wiring and Cable Terminals and Splice Clips.

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

J551 - Measurement of Electromagnetic Radiation from Motor Vehicles and Devices.

J553 - Circuit Breakers.

J560 - Primary and Auxiliary Seven Conductor Electrical Connector for Truck-Trailer Jumper Cable.

J561 - Electric Terminals-Eyelet and Spade Type.

J575 - Tests for Motor Vehicle Lighting Devices & Components.

J576 - Plastic Materials, For Use in Optical Parts Such as Lenses and Reflectors of Motor Vehicle Lighting Devices.

J578 - Color Specification for Electric Signal Lighting Devices.

J682 – Front and Rear Wheel Splash and Stone Throw Protection.

J1128 - Low Tension Primary Cable.

J1292 -Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

AMS-STD-595A- Colors Used in Government Procurement

Applications for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096

2.3 INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO 668- Shipping Containers.

ISO standards may be accessed on the internet at <https://iso.org>.

2.4 NATO (STANAG)

NATO (STANAG) 4009 - Interface Geometry, Tractors and Semi-Trailers detailing the tow vehicle/trailer interface. Reference larger swing radius, 90.5 inches.

NATO standards may be accessed on the internet at <https://nso.nato.int/nso/nsdd/listpromulg.html>.

2.5 OTHER STANDARDS

RP 803-Pre-Service Trailer Inspection.

Published by the Technical and Maintenance Council of the American Trucking Association.

2200 Mill Road, Alexandria, VA 22314.

TMC Recommended Practices may be accessed on the internet at <https://tmconnect.trucking.org>.

2.6 ORDER OF PRECEDENCE

In the event of a conflict between the text of this purchase description and the references cited above, the most stringent requirements, as determined by the government, take precedence. Nothing in this purchase description, however, supersedes applicable state and federal safety laws and regulations.

3.0 GENERAL REQUIREMENTS

Materials shall be as specified herein. When materials are not specified, the trailer and all parts thereof shall be furnished to provide the intended function, durability, safety, and maintain good long-term appearance. All materials shall be new, shall be suitable for the intended purpose, and shall be free of any characteristics or defects in material and workmanship, which may affect the performance, function, durability, and serviceability of the finished trailer, or detract from its appearance. The government reserves the right to make the final determination of the suitability of all components and their arrangement on and in the trailer.

All items shall be of good quality, in safe operating condition. All safety devices shall be fully functional; to include all manual and automatic emergency shut-off devices. All safety guards, warning labels, and safety related instruction labels shall be properly always affixed.

The trailers shall be compatible with the following tractors, Government Owned Vehicles:

Note: EXWC shall briefly describe their commercial and tactical tractors and salient features of each type including:

Make/Model

Mack Granite 6X4, 160K Tractor, 2500 psi, 20 gpm

Series: Mack GR64BTX

GCW: 66,000 Lbs.

GCWR: 176,000 Lbs.

FAWR: 22,000 Lbs. but limited to 20,000 Lbs. due to tires

RAWR for Tandem: 46,001 Lbs. each axle

Wheelbase: Center of front axle to center of tandem spread 204.5"

5th wheel Height: 53.5"

Rear swing/clearance area: Center of 5th wheel: 75"

Front swing/clearance area: Center of 5th wheel from 5th wheel to front most obstruction on cab 66.5" / from 5th wheel to rear most of tractor 75" (with adjustability of 5th wheel left at 6" toward the rear of tractor left and 17.5" inches toward the cab of tractor left.

Tractor wet line kit rating: 2500 psi. @ 20 GPM. With ¾" quick connect coupler 30-gallon reservoir

Heavy Truck Tractor (HTT),

Series: Navistar Heavy Truck Tractor

GCW: 130,000 Lbs.

GCWR: 166,000 Lbs.

FAWR: 20,000 Lbs.

RAWR for Tandem: 84,000 Lbs. each axle.

Wheelbase: Center of Front Axle to center of intermediate axle 197.5" and then from center of front axle to center of most rear axle 258"; spacing between rear duals is 60" front center of intermediate to center of rear dual.

5th wheel Height: 52" – 59" adjustable

Rear swing/clearance area: Center of 5th wheel 73"

Front swing/clearance area: Center of 5th wheel 58" (with adjustability of 5th wheel left at 15.5" toward the rear of tractor left and 30" inches toward the cab of tractor left.)

Tractor wet line kit rating: 3000 psi. @ 57 GPM. With 1 ¼" quick connect coupler

Heavy Interim Truck Tractor (HITT).

Series: International Work Star Model 7600 SBA 6X6.

GCW: 62,000 lbs.

GCWR: 26,280 lbs.

FAWR: 16,000 Lbs.

RAWR for Tandem: 46,000 Lbs each axle.

Wheelbase: Center of front axle to center of tandem spread 217"

5th wheel Height: 50 “- 52”

Rear swing/clearance area: Center of 5th wheel to rear most point on tractor 99” (However 5th wheel is adjustable)

Front swing/clearance area: Center of 5th wheel to forward most point of obstruction 79.5 “(with adjustability of 5th wheel left 41” toward the rear of tractor and 14” inches toward the cab of tractor left)

Tractor wet line kit rating: 13-46 GPM @2500 psi with 1 ¾” female coupler

MTVR Tractor,

Series: MK31

GCW: 62,000 Lbs.

GCWR: 46,000 Lbs.

FAWR: 16,000 Lbs.

RAWR for Tandem: 46,000 Lbs each axle.

Tandem Spread: 56.4 “Centerline to centerline of both rear axle hubs

Wheelbase length: 212.2 “

5th wheel type: Fixed

5TH wheel height: 52” - 59” adjustable suspension.

Rear swing/clearance area: Center of 5th wheel to rear most point on tractor 69”.

Front swing/clearance area: Center of 5th wheel to forward most point of obstruction 62.5”

Tractor wet line kit rating: N/A

3.1 CONFIGURATION REQUIREMENTS

The trailer shall be rated 60 Tons, low bed, drop deck, tri-axle, with hydraulic removable gooseneck (RGN). The trailer shall be used to transport regular, tracked, and oversized construction equipment, construction equipment attachments, and multiple configurations of palletized supplies, and miscellaneous construction supplies.

BODY/FRAME:

Steel.

PAYLOAD CAPACITY

120,000 lbs. minimum.

120,000 lbs. or greater concentrated in 16 ft. or less distributed evenly across the load space.

DIMENSIONS

Overall Length: Not to exceed 53 ft.

Overall Width: Not to exceed 8.5 ft. (102 in.) (Excluding outriggers).

DECK

Composite decking, Rumber or similar, must be impervious to fluids, insects, mud, oil, and UV rays.

Decking to have proper cross member spacing to support decking/load.

Ground clearance: 6 inches or greater when loaded at maximum weight. Should be sufficient to prevent damage to the axles or frame when trailer in operation while loaded.

Fixed, front-loading ramp(s), with welded 3/4" (minimum) cleat bars/traction aids, Weep holes at bottom for drainage.

Removable heavy-duty full-length outriggers.

Pin on front box outriggers with lockout mechanism.

Hard wood planks capable of expanding the width of the deck to a minimum of 10.5 feet (126 inches). All wood/plank information to be submitted to USN for review/approval.

Outrigger lumber shall be secured by using spring actuated flat steel plate clamps.

Centerline deck storage for hardwood outrigger planks.

GOOSENECK

Hydraulic, Removable Gooseneck (RGN).

Swing radius*: 84 In. (Minimum).

Automatic positive lock or pawl.

Five (5) to seven (7) Position variable ride height adjustment lever/controls to maintain a minimum 6-inch ground clearance when towed.

Hydraulic levers on driver side, marked with instruction for operation. Hydraulic Controls, driver side, shall be labeled with data plate above controls for what those functions are and then a separate data plate for the downward controls and posted beneath the controls.

OPTIONAL

GOOSENECK (NATO STANAG 4009*)

When this option is ordered, a gooseneck with a minimum swing clearance of 90.5" shall be provided. The intent and purpose of this option is to assure compatibility with tow tractors in accordance with NATO STANAG 4009.

*Reference: NATO (STANAG 4009) - Interface Geometry, Tractors and Semi-Trailers regarding the tow vehicle/trailer interface and larger swing radius of 90.5 inches.

NATO standards may be accessed on the internet at <https://nso.nato.int/nso/nsdd/listpromulg.html>.

POWER/ENGINE

Pony Motor, Diesel (JP8/F24/High Sulphur) compatible engine shall be protected from elements (E.g., At front/top of gooseneck; with top cover or similar; centered within gooseneck frame; in enclosure; other, etc.).

Mounting platform engine is on must allow access to drain plugs.

Motor shall be easily accessible for maintenance/service.

Motor ignition to have switch or strong tethering mechanism for key to prevent loss.

The pull cord on pony motor must be mounted in a position that allows unimpeded range of motion.

The engine/pump assembly shall be easily accessible for maintenance/service/operation, including access to controls, batteries.

Provisions shall be provided to allow the draining of engine oil.

The engine must allow access to all components of the engine 360 degrees so that operators or maintenance personnel can access all areas of the engine.

Engine exhaust shall be directed away from the operator area and controls located on the side of the trailer.

The battery on pony motor must be located so it is easily accessible with plenty of room to hook up jumper cables if needed or clean terminals etc. without hinderance.

Battery Box with bottom weep holes.

Engine hydraulic pump, tank, and battery must be easily accessible and readily available to operators without the need for tools.

HYDRAULIC SYSTEM

Hydraulic system shall include at a minimum a pump, reservoir, cylinders, valves, lines and connectors. The reservoir shall include sight glass and a dipstick.

Hydraulic fittings shall be Dual hydraulic wet line connectors/couplers (Standard two-line with adapters), Multi-Purpose, ISO 7241 series B Interchange, Female Hydraulic Quick Couplings - 60 Series, 3/4" 14 NPTF, with male/female rubber boots.

The trailer shall be able to operate with a wet line kit supplying a minimum of 12 gallons per minute (GPM) at 2500 pounds per square inch (PSI) and a maximum of 3000 PSI at 57 GPM. The range of capacities is required to allow use of both commercial and tactical Government Owned Vehicles (GOV) as specified.

Auto Diverter Valve for Power Take Off (PTO)/Pony Motor Operations.

The hydraulic system shall be compatible with the following fluids, MIL-PRF-2104J (OE/HDO-10) Lubricating Oil, Engine Grade 10W Oil (used by Navy Deployable Units), and AW32 Hydraulic Oil

The hydraulic system must be compatible with all of the Navy prime movers listed above.

Hydraulic lines shall have a sling available for use to protect lines from getting damaged when the trailer is connecting and disconnecting from tractor/prime mover.

All hydraulic lines will not be taught and must have some slack built into them, so they don't become damaged.

Labels or stenciling shall be provided and installed, indicating the inlet and outlet connections.

Hydraulic tank must have either a sight glass or dipstick that can be read easily.

Stenciling shall be provided as per P-300 and P-480 publications and installed, indicating the types/grades of fuels, oils, and lubricants at each applicable fill location.

Refer to Paint and Marking Section for further guidance/information on markings for Fuel and Hydraulic storage tanks.

ELECTRICAL

6-28-volt DC Electrical / LED lighting system.

NATO intravehicular plug, 12 pin/24V connection.

Commercial intravehicular plug, 7-way, 12V connection

ABS connector for ABS system.

ABS light, driver side rear.

Tail lamps, stop lamps, and reflectors shall be capable of operation throughout all operating conditions and shall Department of Transportation (DOT) standards.

All running lights shall have proper weatherproofing.

All wiring harnesses shall be modular, sealed using weatherproof connectors, enclosed in protected jacket/cover, including grommets to prevent wiring damage, and be secured with hard brackets and corrosion resistant steel bolts and nuts.

All wiring connections shall be waterproof and protected where it terminates at fixtures.

All hoses must be secured to chassis with hard brackets and corrosion resistant steel bolts and nuts not attached to axle.

all electrical to be secured with bolt through clam shell brackets with rubber inserts and enough of them to prevent electrical drooping or movement of electrical while in operation and must be fashioned to trailer in a way that does not allow it to be caught of brush or other surfaces when operation in a off highway scenario; no other electrical securing options such as zip ties, tape, wire, hangers etc. will be accepted.

REAR BRIDGE

Tri-Axle, fixed frame.

Load bearing Rear Bridge with weight evenly distributed over axles.

3rd Axle, pneumatic/air lift (no flip axle).

Rear loading ramps capable of loading up to a minimum of 40,000 pounds over rear axles.

Ramps to have side handles, cradle locking arm with keeper/catch, and clasp pin with tether or similar.

Weep holes at the bottom of the ramps for drainage.

Open center rear trough with heavy-gauge steel floor plates with extended sides and rigid spine.

Trough plates must be easily removable.

D-Rings Center Deck, in well, and on each side of transition point at rear bridge to assist with tying down equipment.

Minimum ¾ inch welded Cleat Bars/traction aids at transition points (main deck to rear bridge and rear bridge to beavertail).

SUSPENSION/AXLES

Suspension dump valve (Hold, raise, lower with auto dump), driver side between rear axles. Language to be included in the Operator's manual.

Laser Engraved metal plate or decal located on the gooseneck stating: "Dump suspension before disconnect" or similar.

Automatic slack adjusters.

Mechanical Hub-odometer mounted on driver's side front axle.

Oil bath hubs with visible oil level indicators.

BRAKES

Anti-Lock Braking System (ABS), Commercial Standard. ABS system/wiring should be adequately protected and mounted so as not to interfere with prime mover connection or operation of the trailer.

Mechanical air tank draining control, easily accessible from side or rear.

TIRES

New, non-retread, Commercial Highway, Pneumatic Rubber radial tires, rated for trailer at full load weight capacity.

All tires must be same, brand matching.

Tire size information shall be provided.

Manufacturer date on all tires shall be no greater than one year at time of delivery of equipment. (No double coins).

Valve extensions shall be provided on inner and outer tires and must be easily accessible

Refer to Paint and Marking Section for further guidance/information.

WHEELS

Solid steel single piece wheels with lugs, dual wheels (super singles are prohibited).

Wheel/Rim size information shall be provided.

All wheels used must have easy access to air valves in order to air up or down, additionally only analog hub meters will be used.

Refer to Paint and Marking Section for further guidance/information.

3.2. PERFORMANCE

Trailer shall be: Commercial design, conforming to 50 State Department of Transportation (DOT) requirements and Federal highway and bridge laws.

Exception: Permitting may be required for transport in certain U.S. states and or across state lines.

Operable cross country on highways, paved secondary, gravel, unimproved, or cobblestone roadways.

Capable of operating in a wide variety of climatic conditions and environments, able to function in operating temperatures ranging from -25 degrees to 130 degrees Fahrenheit. Must be capable of 10 hours run time continuously minus refueling operations with an hour of maintenance between periods.

Air, brake, hydraulic lines, and fittings shall be free from leaks and mounted to prevent damage during vehicle operation.

All welding to include welded, bolted, and riveted construction utilized, shall be in accordance with the highest standards of industry and must conform to Structural Welding Code for Steel and Stainless Steel IAW American Society of Mechanical Engineers (ASME) Section IX and American Welding Society (AWS) Standards.

All welds shall be free of excess slag and splatter.

All welds shall be uniform, have adequate penetration, and be free of voids.

All rubber components shall be ozone and UV resistant.

AUXILIARY

Glad hand covers shall be affixed with chain to glad hands., covers shall be positioned to not interfere with coupling to tractor and interfering with tractor while hooking and unhooking or turning while in operation

D-Rings (Tie Down) for Ship and Air transport shall be in accordance with MIL-STD 209K and MIL-STD-1791D_CHG-1 requirements.

D-rings, eight (8) Minimum per side of frame, plus two (2) on each side of gooseneck, 10,000 pounds rating or greater each. D-Rings must be sufficient for load and welded properly.

D Rings, eight (8) welded down center frame for Equipment Tie Down, rings must be sufficient for load and welded properly.

Two sealed, weatherproof, lockable toolboxes/storage compartments with bottom weep holes, lockable lid latches, and reinforced flush fitted door hinge(s) to hold chains, binder, electrical pigtails, cargo straps, tools. (Position of boxes to be recommended by OEM and to be approved by USN).

One (1) new spare tire/wheel to match other tires/wheels on trailer; Mounted spare should not interfere with prime mover connection, loading to aircraft or ship via ramp, lift or tie down, or operation/loading of trailer.

½ inch Chain/Key slots on both right and left outer structural beam rails.

½ inch 4-way center Chain/Key slots to complement outer chain key slots.

Wide load signs with mounting brackets.

Flag holders (2 pair); one (1) pair each at front and back of trailer.

Tools: to include breaker bar, lug wrench, socket tool, to remove wheels.

Heavy Duty Black rubber mud guards without markings.

Reflective Tape to meet NHTSA FMVSS 108 DOT C2.

MILITARY MODIFICATIONS

The trailer shall be able to interface with standard prime mover models with **52-to-57-inch** 5th wheel height.

3.3. AIR TRANSPORT:

The trailer shall be capable of air transport by C-17, and C-5 Aircraft. Refer to MIL-STD-1366 and MIL-STD-1791D_CHG-1.

Transportability shall be maximum curb weight of trailer.

4540.07 CHG-3 Operation of the DoD Engineering for Transportability and Deploy ability Program.

MIL STD-1366E Interface Standard for Transportability Criteria

MIL-STD-1791D_CHG-1 Designing For Internal Aerial Delivery in Fixed Wing Aircraft.

MIL-STD-209K Interface Standards for Lifting and Tie down Provisions

Process for ATTILA Air Transportability Certification

Equipment or ground vehicles that require USAF Fixed Wing Air Transportability certification via C130E/H/J, C17, or C5 shall be in accordance with the following:

DODI 4540.07 CHG-3 Operation of the DoD Engineering for Transportability and Deploy ability Program.

MIL STD-1366E Interface Standard for Transportability Criteria

MIL-STD-1791D_CHG-1 Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

MIL-STD-209K Interface Standards for Lifting and Tie down Provisions

The following sequence of events shall occur:

The General Services Agency (GSA), Defense Logistics Agency (DLA), or other Acquisition source will:

Notify the purchasing customer/Government Agency sponsor of contract award.

Forward a copy of the GSA, DLA, or other Acquisition source award order/contract to the purchasing customer/Government Agency and awarded OEM vendor/contractor.

a) (GSA, DLA, or other Acquisition Source) Order number or contract to include all contract options ordered

b) Copy of the Motor Vehicle Delivery Order (MVDO) (as applicable)

c) Purchasing customer/Government Agency Sponsor contact information (name/phone/email).

d) OEM vendors / contractors contact information (name/phone/email)

The purchasing customer/Government Agency Sponsor will:

Act as the OEM vendor/contractor's Government Agency Sponsor to submit an ATTILA package for any equipment or ground vehicles that require Air Transportability certification.

Submit a Sponsor Letter **within 10 working days** post contract award to request to have the Air Transportability Test Loading Agency (ATTILA) open a project for an Air Transportability Certificate.

Sponsor requests to be sent by the purchasing customer/Government Agency to the

Air Transportability Test Loading Agency (ATTILA) at Wright-Patterson Air Force Base at (ATTILA@us.af.mil). Provide ATTILA with the following information as reference for the ATTILA project case file:

- a) Product: (insert Product name here i.e. Truck Tractor 20T, Trailer Low Bed RGN 60T, etc.)
- b) Contract: (GSA, DLA, or other Acquisition Source) Order number to include all contract options ordered
- c) Copy of the Motor Vehicle Delivery Order (MVDO) (as applicable)
- d) Purchasing customer/Government Agency Sponsor contact information (name/phone/email).
- e) OEM vendors / contractors contact information (name/phone/email)

Provide awarded OEM vendor/contractor with a checklist of items required by ATTILA that they **must complete and submit to** the purchasing customer/Government Agency upon completion of the product build as part of the ATTILA Transportability package submission.

Confirm receipt of OEM vendor/contractor ATTILA package submittal to purchasing customer/Government Agency Sponsor.

Verify final package submitted to ATTILA is complete including all required information.

Track status of the ATTILA certification review including the date of first submittal, any requests for information by ATTILA, any vendor resolution of nonconformance to meet ATTILA requirements, or other items as required.

Forward copies of final ATTILA certification to EXWC EX branches (EX2/EX4/EX5 as applicable)

During production, the purchasing customer/Government Agency Sponsor shall:

Collaborate with OEM vendor/contractor to review work in progress: the OEM Finite Element Analysis (FEA) and other engineering data, drawings, photos, etc.

The following angles, dimensions, and descriptions should be checked against data and drawings (see ATTILA Universal Data Spreadsheet for instructions/information) as part of the ATTILA technical data package submittals:

- a) Angle of approach
- b) Ramp breakover angle
- c) Angle of departure
- d) Height, longitudinal location and identification of highest component on the equipment/vehicle
- e) Dimensions and locations of any significant projections on truck
- f) Curb weight of each axle
- g) Wheelbase
- h) Front overhang
- i) Rear overhang
- j) Articulation of rear suspension, unloaded, each axle (curb weight)
- k) Rear axle spacing l) Axle rating, front, and comparison to 1 ¼ times (curb weight) load
- m) Axle rating, rear, and comparison to 1 ¼ times (curb weight) load
- n) Suspension rating, front, and comparison to 1 ¼ times (curb weight) load
- o) Suspension rating, rear, and comparison to 1 ¼ times (curb weight) load
- p) If axle stops are to be removed for ramp loading on aircraft, verification that the driveline remains intact when creating maximum ramp slope. Provide guidance and recommendations on changes needed to correct any issues identified where the certification request may be denied, or the information package is incomplete.
- q) Configuration and directional capability of equipment/ground vehicle tie down provisions (Reference MIL-STD-1791-D_CHG1, 5.3.3.3.2)

Delivered/ Fielded Equipment with NO ATTILA Certification:

In cases where Equipment or Ground Vehicles were delivered to a customer without an Air Certification letter, the purchasing customer/Government Agency Sponsor shall provide the OEM vendor/contractor with guidance on the ATTILA process.

Working with the Acquisition Source (GSA, DLA, other) the purchasing customer/Government Agency Sponsor will direct the OEM to either retrieve the Equipment / Ground Vehicle delivered for needed modifications or arrange for a local shop to perform any needed modifications to satisfy ATTILA requirements to obtain transportability certification. Upon completion of modification work, the OEM vendor/contractor shall advise the purchasing customer/Government Agency Sponsor.

The Purchasing customer/Government Agency Sponsor will review OEM corrections of any issues and then submit a technical data package on behalf of the OEM/vendor for ATTLA review for certification.

Air Transportability Test Loading Agency (ATTLA) will:

Contact the purchasing customer/Government Agency Sponsor to acknowledge request and to confirm an ATTLA project has been opened/established. Other information/data/analysis could be required.

(Pending action by the purchasing customer/Government Agency to submit the vendor/contractor's completed package for ATTLA review, analysis, and certification of for Air Transportability).

Upon receiving the submission of the data package (i.e. characteristics of the equipment, 3-view dimensional drawings, proposed tiedown pattern) ATTLA will then conduct a technical analysis of the transportability at no cost to the requesting federal agency.

ATTLA will compare the OEM vendor/contractor data and all weight estimates against the rated capacities (E.g. Equipment, axle, suspension, etc.) to determine compliance with MIL-STD-1791D_CHG-1 Department of Defense Interface Standard for Internal Aerial Delivery in Fixed Wing Aircraft.

Should submitted package requirements be incomplete, or should an item fail to qualify for airlift transport, ATTLA will provide an explanation stating reasons for rejection to the purchasing customer/Government Agency and make recommendations for changes that will allow an item to meet ATTLA certification requirements.

Upon approval, ATTLA will provide a transportability certification memo to the purchasing customer/Government Agency sponsor stating the conditions of approval / certification of transportability within 60 days of receiving a complete package with all proper requirements met.

Note: For items in development, ATTLA **may** offer design assistance through the applicable procurement office to ensure that the vendor produces an item that ultimately meets all airlift objectives. Because developmental items are subject to continual changes, ATTLA personnel may wish to participate in design reviews and other technical interchanges during the evolution of a design. However, funding must be provided for ATTLA personnel to attend onsite activities at an OEM vendor production site.

Teams or video teleconference capability may be offered by ATTLA at no charge to the purchasing customer/Government Agency or OEM at any phase of equipment/vehicle production. **(Meetings must be requested/scheduled with ATTLA)**

The OEM vendor/ contractor will:

Upon completion of the build of the equipment/vehicle, submit the following technical data as a package to the Mailing Address of the purchasing customer/Government Agency Sponsor (as provided on the Motor Vehicle Delivery Order (MVDO) or contract, or other mailing address identified/specified by the purchasing customer).

ATTLA Universal Data Spreadsheet (appropriate tab sheets completed - **See Instructions and General Information tab sheet for guidance on completion**)

- a) Identify equipment type needing Air Transportability Certification.
- b) Name of the item from the data plate (E.g. crane, tracked vehicle, truck tractor, trailer, etc., with naming format as: Truck Stake 2-Door 20 Foot Bed 30T, Trailer Lowbed HYD RGN 60T, etc.
- c) Identify type of USAF aircraft (C-130E/H/J, C-17, or C-5) (provided by the purchasing customer/government agency sponsor). ATTILA will identify the USAF aircraft required if different from the submittal package request.
- d) The purchasing customer/government agency sponsor will provide the OEM/vendor with a technical data package template recommended as a tool/ resource to ensure all requirements are provided in the submittal for transportability certification. This template will help the OEM identify MIL-STDs and requirements needed for the specific variant of equipment or vehicle to be certified.
- e) Provide a technical data package as per MIL-STD-1791-D para-6.4 Air Transport Certification Process. Package should include physical characteristics of the equipment, structural strength with substantiating information (for analysis of tiedown capability), and proposed tiedown pattern. Dimensions may be submitted using drawings, certain CAD files, the ATTILA data sheet, or annotated photographs. The structural analysis should list assumptions, material properties, and force diagrams in addition to calculations and conclusions.
- f) The contractor shall supply the ATTILA Approved Certification memo to the purchasing customer/Government agency after receipt of the first production equipment/vehicle.

A copy of the ATTILA certification memo shall be forwarded to the GSA IOA when it is available.

The contractor shall present to the GSA IOA a certified weight ticket of the completed trailer including curb weights of all axles. The GSA IOA shall then confirm and validate that the actual loaded axle weights are equal to or less than 80% of their ratings in accordance to the ATTILA certification. ATTILA (The Air Transportability Test Loading Agency) is located at Wright Patterson Air Force Base and can be contacted via e-mail at (ATTILA@us.af.mil).

The equipment/ground vehicle shall be inspected to determine that it conforms to the approved certification memo the contractor received from Air Transportability Test Loading Agency (ATTILA).

Lift and Cargo/Equipment Tie Down Provisions:

The contractor shall be required to provide engineering analysis or other documentation, to include design data plate with specifications and drawings mounted to trailer, to validate Cargo and Equipment tie down and lift provision testing and certification to include maximum Cargo and Equipment tie down and lifting/sling weight capacity in accordance with MIL-STD-209K.

Pull Test:

If analysis alone cannot positively determine the transportability of the Equipment / Ground Vehicle, then a test loading may be required.

Test loadings are done when the transportability of an item is uncertain.

Test loads require a formal test report and are usually conducted as Special Airlift or Transport Missions.

The purchasing customer/Government Agency Sponsor shall be responsible for requesting a test loading as well as all costs associated when it is determined to be required.

3.4. Military Surface Deployment and Distribution Command Transportation Engineering Agency (SDDCTEA)**Engineering structural analysis.**

Prior to all testing, the OEM Vendor/contractor may provide dimensional and material design data on the provisions and their surrounding structure. This data will be used to perform a structural analysis of the provisions. In lieu of providing the required design information, the contractor can perform a structural analysis and provide the results of the analysis for review and concurrence. Either the required data or the contractor's structural analysis results should be provided at least **90 days** prior to scheduled provision testing. The purpose of the structural analysis is to assess the risk of testing, thereby reducing the number of failures or possibly eliminating the test in acceptable cases. This will result in lower overall acquisition costs to materiel developers.

Removal or relocation of mechanically attached (non-welded, non-riveted, etc.) components with common tools, requiring not more than 1 man-hour total to remove, relocate and tie down, and not more than 1 man-hour total to return the vehicle to its original, as opposed to reduced, configuration; shall be acceptable.

- a. The self-mobility of the vehicle shall not be affected by reducing its configuration. Tie downs for removed or relocated equipment shall be furnished.
- b. For Trailers, the rated capacity of the axles and suspension system shall be not less than 1.25 times the load imposed on each by the curb weight of the trailer.

- c. The vehicle shall be transportable as described above without any other special provisions and without any shoring unless unavoidable without significant alteration to the vehicle.
- d. The vehicle shall not be delivered to the Government in its reduced configuration.

For Rail or pull test certification contact: the Military Surface Deployment and Distribution Command Transportation Engineering Agency (SDDCTEA) for modes of transportability.

SDDCTEA is in Newport News, VA, and can be contacted by e-mail at “dpemail@tea.army.mil” or by phone at (757) 599-1113.

A transportability report form for SDDCTEA can be obtained at <http://www.tea.army.mil/pubs/nr/deploy/transinstruction/DI-PACK-80880B.pdf> \t “_new”. References, publications, assistance and engineering services are available from SDDCTEA. Testing information can be obtained from the Aberdeen Test Center (ATC). ATC can be contacted through their website at “<http://www.atc.army.mil>

3.5. LIFT AND TIE DOWN PROVISIONS:

The trailer shall be equipped with lift and tie down provisions to ensure interoperability between transported equipment and lifting and tie down devices commonly used in the transportation environment. If there is any drainage holes utilized in pockets of tie down or lift ring areas will require protection from chains or other foreign objects not part of the trailer from striking anything but pocket and lift ring area, reason is to prevent any damage to other parts of trailer, i.e. Valve bodies, hoses, electrical, lights etc.

Lifting:

Lift provisions (such that no spreader bar is required) shall meet the Interface Standards and requirements for transportability in accordance with MIL-STD-1366, and shall permit lifting of the trailer, without payload, in accordance with MIL-STD-209K.

Total lift capacity, to include maximum sling lift weight, shall be reflected via data plate.

All design loads for lift provisions shall be for maximum curb weight of trailer.

Total Lift Ring “Rating” for each ring shall be reflected via data plate and on the D-Ring. The capacity of each tie down D-Ring shall be a Raised Casting located on the body of the D-Ring and mounted so that the capacity information is facing outward for visibility.

All lift provisions must be seated, properly welded to frame, and rated for capacity of load. and if there is any drainage holes utilized in pockets that they have protection from chains or other foreign objects not part of the trailer from striking anything but pocket and lift ring.

Refer to Paint and Marking Section for further guidance/information.

Tie Down:

Tie Down provisions and hardware shall meet the Interface Standards and requirements for transportability in accordance with MIL-STD-1366 to permit transport of the trailer, with or without payload, in accordance with MIL-STD-209K and MIL-STD-1791D CHG-1.

All design loads for tie down shall be for full load capacity of trailer.

Total Tie Down “Rating” for each ring shall be reflected via data plate and on the D-Ring. The capacity of each tie down D-Ring shall be a Raised Casting located on the body of the D-Ring and mounted so that the capacity information is facing outward for visibility.

All Tie Down provisions must be seated, properly welded to frame, and rated for capacity of load.

The contractor shall be required to provide engineering analysis or other documentation, to include design data plate with specifications and drawings mounted to trailer, to validate tie down and lift provision testing and certification to include maximum tie down and lifting/sling weight capacity in accordance with MIL-STD-209K.

A transportation/shipping data plate shall be furnished, secured to the unit with screws, rivets, or other mechanical devices in accordance with MIL-STD-209K, and shall conform to composition A (class 1 or 2) or composition C of A-A-50271. The shipping data plate shall be inscribed or etched, with a diagram showing the lifting points/attachments and lifting slings, capacity of each lift point/attachment, maximum sling lift capacity, and the required length and size of each sling cable. A silhouette of the trailer showing the center of gravity shall be provided on the transportation/shipping plate.

Data plates/stenciled/painted markings shall identify all lifting and tie down points/attachments and capacity. Alternative use of decals by OEM in place of stencil/paint must be submitted for review/approval by USN.

Refer to Paint and Marking Section for further guidance/information.

ANALYSIS:

The contractor shall perform a structural analysis of the tie down and lifting provisions and the surrounding structural elements in accordance with MIL-STD 209 requirements. In cases when the structural analysis indicates the provisions will clearly pass the requirements, actual physical testing may not be necessary. In cases where the structural analysis indicates the provisions will marginally pass the requirements, redesign or testing shall be required of the contractor. In cases where the structural analysis indicates the provisions will clearly fail the requirements, a redesign of the provisions shall be required.

The contractor shall be required to provide engineering analysis or other documentation, to include design data plate with specifications and drawings mounted to trailer, to validate tie down and lift provision testing and certification to include maximum tie down and lifting/sling weight capacity in accordance with MIL-STD-209K.

A transportation/shipping data plate shall be furnished, secured to the unit with screws, rivets, or other mechanical devices in accordance with MIL-STD-209K, and shall conform to composition A (class 1 or 2) or composition C of A-A-50271. The shipping data plate shall be inscribed with a diagram showing the lifting points/attachments and lifting slings, capacity of each lift point/attachment, maximum sling lift capacity, and the required length and size of each sling cable. A silhouette of the trailer showing the center of gravity shall be provided on the transportation/shipping plate.

Data plates/stenciled/painted markings shall identify all lift and tie down points/attachments and capacity. Alternative use of decals, laser engraved, riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN.

3.6. AIR TRANSPORT:

The trailer shall be capable of air transport by C-17, and C-5 Aircraft. Refer to MIL-STD-1366.

Refer to MIL-STD-1366 and DODI 4540.07 CHG-3 Operation of the DoD Engineering for Transportability and Deploy ability Program.

MIL STD-1366E Interface Standard for Transportability Criteria

MIL-STD-1791D_CHG-1 Designing For Internal Aerial Delivery in Fixed Wing Aircraft.

MIL-STD-209K Interface Standards for Lifting and Tie down Provisions

Process for ATTLA Air Transportability Certification

Equipment or ground vehicles that require USAF Fixed Wing Air Transportability certification via C130E/H/J, C17, or C5 shall be in accordance with the following:

DODI 4540.07 CHG-3 Operation of the DoD Engineering for Transportability and Deploy ability Program.

MIL STD-1366E Interface Standard for Transportability Criteria

MIL-STD-1791D_CHG-1 Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

MIL-STD-209K Interface Standards for Lifting and Tie down Provisions

The following sequence of events shall occur:

The General Services Agency (GSA), Defense Logistics Agency (DLA), or other Acquisition source will:

Notify the purchasing customer/Government Agency sponsor of contract award.

Forward a copy of the GSA, DLA, or other Acquisition source award order/contract to the purchasing customer/Government Agency and awarded OEM vendor/contractor.

- a) (GSA, DLA, or other Acquisition Source) Order number or contract to include all contract options ordered
- b) Copy of the Motor Vehicle Delivery Order (MVDO) (as applicable)
- c) Purchasing customer/Government Agency Sponsor contact information (name/phone/email).
- d) OEM vendors / contractors contact information (name/phone/email)

The purchasing customer/Government Agency Sponsor will:

Act as the OEM vendor/contractor's Government Agency Sponsor to submit an ATTILA package for any equipment or ground vehicles that require Air Transportability certification.

Submit a Sponsor Letter **within 10 working days** post contract award to request to have the Air Transportability Test Loading Agency (ATTILA) open a project for an Air Transportability Certificate.

Sponsor requests to be sent by the purchasing customer/Government Agency to the

Air Transportability Test Loading Agency (ATTILA) at Wright-Patterson Air Force Base at (ATTILA@us.af.mil). Provide ATTILA with the following information as reference for the ATTILA project case file:

- a) Product: (insert Product name here i.e. Truck Tractor 20T, Trailer Low Bed RGN 60T, etc.)
- b) Contract: (GSA, DLA, or other Acquisition Source) Order number to include all contract options ordered
- c) Copy of the Motor Vehicle Delivery Order (MVDO) (as applicable)
- d) Purchasing customer/Government Agency Sponsor contact information (name/phone/email).
- e) OEM vendors / contractors contact information (name/phone/email)

Provide awarded OEM vendor/contractor a checklist of items required by ATTILA that they **must complete and submit to** the purchasing customer/Government Agency upon completion of the product build as part of the ATTILA Transportability package submission.

Confirm receipt of OEM vendor/contractor ATTILA package submittal to purchasing customer/Government Agency Sponsor.

Verify final package submitted to ATTILA is complete including all required information.

Track status of the ATTILA certification review including the date of first submittal, any requests for information by ATTILA, any vendor resolution of nonconformance to meet ATTILA requirements, or other items as required.

Forward copies of final ATTILA certification to EXWC EX branches (EX2/EX4/EX5 as applicable)

During production, the purchasing customer/Government Agency Sponsor shall:

Collaborate with OEM vendor/contractor to review work in progress: the OEM Finite Element Analysis (FEA) and other engineering data, drawings, photos, etc.

The following angles, dimensions, and descriptions should be checked against data and drawings (see ATTILA Universal Data Spreadsheet for instructions/information) as part of the ATTILA technical data package submittals:

- a) Angle of approach
- b) Ramp breakover angle
- c) Angle of departure
- d) Height, longitudinal location and identification of highest component on the equipment/vehicle
- e) Dimensions and locations of any significant projections on truck
- f) Curb weight of each axle
- g) Wheelbase
- h) Front overhang
- i) Rear overhang
- j) Articulation of rear suspension, unloaded, each axle (curb weight)
- k) Rear axle spacing l) Axle rating, front, and comparison to 1 ¼ times (curb weight) load
- m) Axle rating, rear, and comparison to 1 ¼ times (curb weight) load

- n) Suspension rating, front, and comparison to 1 ¼ times (curb weight) load
- o) Suspension rating, rear, and comparison to 1 ¼ times (curb weight) load
- p) If axle stops are to be removed for ramp loading on aircraft, verification that the driveline remains intact when cresting maximum ramp slope. Provide guidance and recommendations on changes needed to correct any issues identified where the certification request may be denied, or the information package is incomplete.
- q) Configuration and directional capability of equipment/ground vehicle tie down provisions (Reference MIL-STD-1791-D_CHG1, 5.3.3.3.2)

Delivered/ Fielded Equipment with NO ATTILA Certification:

In cases where Equipment or Ground Vehicles were delivered to a customer without an Air Certification letter, the purchasing customer/Government Agency Sponsor shall provide the OEM vendor/contractor with guidance on the ATTILA process. Working with the Acquisition Source (GSA, DLA, other) the purchasing customer/Government Agency Sponsor will direct the OEM to either retrieve the Equipment / Ground Vehicle delivered for needed modifications or arrange for a local shop to perform any needed modifications to satisfy ATTILA requirements to obtain transportability certification. Upon completion of modification work, the OEM vendor/contractor shall advise the purchasing customer/Government Agency Sponsor. The Purchasing customer/Government Agency Sponsor will review OEM corrections of any issues and then submit a technical data package on behalf of the OEM/vendor for ATTILA review for certification.

Air Transportability Test Loading Agency (ATTILA) will:

Contact the purchasing customer/Government Agency Sponsor to acknowledge request and to confirm an ATTILA project has been opened/established. Other information/data/analysis could be required.

(Pending action by the purchasing customer/Government Agency to submit the vendor/contractor's completed package for ATTILA review, analysis, and certification of for Air Transportability).

Upon receiving the submission of the data package (i.e. characteristics of the equipment, 3-view dimensional drawings, proposed tiedown pattern) ATTILA will then conduct a technical analysis of the transportability at no cost to the requesting federal agency.

ATTILA will compare the OEM vendor/contractor data and all weight estimates against the rated capacities (E.g. Equipment, axle, suspension, etc.) to determine compliance with MIL-STD-1791D_CHG-1 Department of Defense Interface Standard for Internal Aerial Delivery in Fixed Wing Aircraft.

Should submitted package requirements be incomplete, or should an item fail to qualify for airlift transport, ATTILA will provide an explanation stating reasons for rejection to the purchasing customer/Government Agency and make recommendations for changes that will allow an item to meet ATTILA certification requirements.

Upon approval, ATTILA will provide a transportability certification memo to the purchasing customer/Government Agency sponsor stating the conditions of approval / certification of transportability within 60 days of receiving a complete package with all proper requirements met.

Note: For items in development, ATTILA **may** offer design assistance through the applicable procurement office to ensure that the vendor produces an item that ultimately meets all airlift objectives. Because developmental items are subject to continual changes, ATTILA personnel may wish to participate in design reviews and other technical interchanges during the evolution of a design. However, funding must be provided for ATTILA personnel to attend onsite activities at an OEM vendor production site. Teams or video teleconference capability may be offered by ATTILA at no charge to the purchasing customer/Government Agency or OEM at any phase of equipment/vehicle production. **(Meetings must be requested/scheduled with ATTILA)**

The OEM vendor/ contractor will:

Upon completion of the build of the equipment/vehicle, submit the following technical data as a package to the Mailing Address of the purchasing customer/Government Agency Sponsor (as provided on the Motor Vehicle Delivery Order (MVDO) or contract, or other mailing address identified/specified by the purchasing customer).

- a) ATTILA Universal Data Spreadsheet (appropriate tab sheets completed - **See Instructions and General Information tab sheet for guidance on completion**)
- b) Identify equipment type needing Air Transportability Certification.
- c) Name of the item from the data plate (E.g. crane, tracked vehicle, truck tractor, trailer, etc., with naming format as: Truck Stake 2-Door 20 Foot Bed 30T, Trailer Lowbed HYD RGN 60T, etc.
- d) Identify type of USAF aircraft (C-130E/H/J, C-17, or C-5) (provided by the purchasing customer/government agency sponsor). ATTILA will identify the USAF aircraft required if different from the submittal package request.
- e) The purchasing customer/government agency sponsor will provide the OEM/vendor with a technical data package template recommended as a tool/ resource to ensure all requirements are provided in the submittal for transportability certification. This template will help the OEM identify MIL-STDs and requirements needed for the specific variant of equipment or vehicle to be certified.

- f) Provide a technical data package as per MIL-STD-1791-D para-6.4 Air Transport Certification Process. Package should include physical characteristics of the equipment, structural strength with substantiating information (for analysis of tiedown capability), and proposed tiedown pattern. Dimensions may be submitted using drawings, certain CAD files, the ATTILA data sheet, or annotated photographs. The structural analysis should list assumptions, material properties, and force diagrams in addition to calculations and conclusions.
- g) The contractor shall supply the ATTILA Approved Certification memo to the purchasing customer/Government agency after receipt of the first production equipment/vehicle.
- h) A copy of the ATTILA certification memo shall be forwarded to the GSA IOA when it is available.
- i) The contractor shall present to the GSA IOA a certified weight ticket of the completed trailer including curb weights of all axles. The GSA IOA shall then confirm and validate that the actual loaded axle weights are equal to or less than 80% of their ratings in accordance to the ATTILA certification. ATTILA (The Air Transportability Test Loading Agency) is located at Wright Patterson Air Force Base and can be contacted via e-mail at (ATTILA@us.af.mil).
- j) The equipment/ground vehicle shall be inspected to determine that it conforms to the approved certification memo the contractor received from Air Transportability Test Loading Agency (ATTILA).

Lift and Cargo/Equipment Tie Down Provisions:

The contractor shall be required to provide engineering analysis or other documentation, to include design data plate with specifications and drawings mounted to trailer, to validate Cargo and Equipment tie down and lift provision testing and certification to include maximum Cargo and Equipment tie down and lifting/sling weight capacity in accordance with MIL-STD-209K.

Pull Test:

If analysis alone cannot positively determine the transportability of the Equipment / Ground Vehicle, then a test loading may be required.

Test loadings are done when the transportability of an item is uncertain.

Test loads require a formal test report and are usually conducted as Special Airlift or Transport Missions.

The purchasing customer/Government Agency Sponsor shall be responsible for requesting a test loading as well as all costs associated when it is determined to be required.

SDDCTEA

Engineering structural analysis.

Prior to all testing, the OEM Vendor/contractor may provide dimensional and material design data on the provisions and their surrounding structure. This data will be used to perform a structural analysis of the provisions. In lieu of providing the required design information, the contractor can perform a structural analysis and provide the results of the analysis for review and concurrence. Either the required data or the contractor's structural analysis results should be provided at least **90 days** prior to scheduled provision testing. The purpose of the structural analysis is to assess the risk of testing, thereby reducing the number of failures or possibly eliminating the test in acceptable cases. This will result in lower overall acquisition costs to materiel developers.

Removal or relocation of mechanically attached (non-welded, non-riveted, etc.) components with common tools, requiring not more than 1 man-hour total to remove, relocate and tie down, and not more than 1 man-hour total to return the vehicle to its original, as opposed to reduced, configuration; shall be acceptable.

- a) The self-mobility of the vehicle shall not be affected by reducing its configuration. Tie downs for removed or relocated equipment shall be furnished.
- b) For Trailers, the rated capacity of the axles and suspension system shall be not less than 1.25 times the load imposed on each by the curb weight of the trailer.
- c) The vehicle shall be transportable as described above without any other special provisions and without any shoring unless unavoidable without significant alteration to the vehicle.
- d) The vehicle shall not be delivered to the Government in its reduced configuration.

For Rail or pull test certification contact: the Military Surface Deployment and Distribution Command Transportation Engineering Agency (SDDCTEA) for modes of transportability.

SDDCTEA is in Newport News, VA, and can be contacted by e-mail at "dpemail@tea.army.mil" or by phone at (757) 599-1113.

A transportability report form for SDDCTEA can be obtained at <http://www.tea.army.mil/pubs/nr/deploy/transinstruction/DI-PACK-80880B.pdf> \t "_new". References, publications, assistance and engineering services are available from SDDCTEA. Testing information can be obtained from the Aberdeen Test Center (ATC). ATC can be contacted through their website at "<http://www.atc.army.mil>

3.7. PAINT AND MARKING

PRIMER, PAINT, FINISH, CORROSION CONTROL

Preparation:

Trailer to be primed/painted **prior** to installation of air, decking, electrical, and hydraulic lines.

Steel; Rumber or composite; or combination of decking shall be installed post paint/coatings so that no unsightly paint or coating overspray is evident on any portion of the decking material.

Chassis frame, undercarriage, underbody, and all base materials (E.g. I-beams, axles, wheels, etc.) should be free of cracks, burrs, sharp edges, and any weld slag, splatter, or rust that may affect the corrosion performance and coatings adhesion to frame, undercarriage, and wash down areas. Must not have any drill through and driving of fasteners into cross members weakening them and leaving metal exposed.

Before painting, all metal surfaces shall be cleaned to remove any metal welding slag, dirt, grease, oil, moisture, silicone, rust, sanding residue, scale, wax, or other foreign matter that may have accumulated.

Exposed fasteners shall be painted flat lusterless green.

Primer/Base:

Two-part Epoxy NON-CARC corrosion resistant coat or zinc rich primer shall be **> or = to 4 MILS Minimum** dry-film thickness (DFT).

Paint Color: Flat lusterless green #34094, in accordance with SAE AMS-STD-595A. (Matte, no gloss).

Finish: Poly Urethane NON-CARC corrosion resistant coat shall be **> or = to 4 MILS Minimum** dry-film thickness (DFT).

Finish shall be suitable for outdoor/UV exposure without undue fading of colors or pigment or chalking.

Combined total:

Primer, paint, and finish coats shall be **no less than 8 MILS Minimum** dry-film thickness (DFT).

A non-skid application (E.g., Applied Grit coating, Grip tape, other) should be applied to top of all horizontal frame surfaces for safety.

All Product information to be submitted to USN for review/approval including potential non-application due to primer/topcoat system.

Undercoat Corrosion Protection:

A corrosion resistance and wear protection preventive undercoating to prevent corrosion.

(E.g., Tectyl, LINE-X, or equivalent) shall be applied evenly to top flanges and outer webs, and all surfaces, frame down, including wash down areas, for protection from corrosion and road debris. Undercoat shall be dry to touch (not tacky), and reflect no excessive running, sagging, or build-up. Any overspray shall be removed.

All Product information to be submitted to USN for review/approval.

VENDOR SUITABLE PRIMER/PAINT/FINISH/UNDERCOAT ALTERNATIVES:

Vendors may propose **ALTERNATIVE** primer/paint/finish and undercoat applications to the USN.

Manufacturers must supply the USN with product chip samples and product technical data sheets (TDS) for primer, paint, finish, and undercoat coatings, (to include product numbers, application, viscosity, certifications/testing documentation (E.g. salt spray, gravel, hardness, etc.), and MIL thickness minimums for review and approval by the USN prior to OEM application.

Alternate coatings shall meet or exceed performance requirements stated above, including:

- (1) Be capable of withstanding 240 hours of salt spray in accordance with ASTM B117/D1654 without rusting or blistering on un-scribed area and rust creepage rating of 5 or higher on scribed specimen
- (2) Chipping resistance in accordance with ASTM D3170.

Suitable primer/paint/finish and undercoat coating alternatives must be compatible and shall be field repairable.

A non-skid application (E.g., Applied Grit coating, Grip tape, other) should be applied to top of all horizontal frame surfaces for safety.

All Product information to be submitted to USN for review/approval including potential non-application due to primer/topcoat system.

REGISTRATION/MARKINGS/LABELS:

U.S. Navy stenciled/painted identification and registration markings shall be matte black and shall be applied to all trailers IAW NAVFAC Requirements consisting of:

- Agency identification, "U.S. NAVY".
- U.S. Navy registration number assigned (E.g., U.S. NAVY XX-XXXXX).

Alternative use of decals, laser engraved, riveted, or welded plates by OEM in place of stencil/paint **must be submitted** for review/approval by USN for **all** markings.



The Government will provide a listing of the USN Registration Numbers to be applied to sides and rear of all trailers.

Markings shall be in the following Font, Format, Size, and Location:

Side:

Centered on each side of the front quarter of the trailer (E.g., gooseneck, side, other).

- Font Format: Stencil.
- Size and Stroke: 3-inches high, ½-inch stroke.
- Dash: ½-inch high, 1-inch long.
- Color: Matte Black.
- Example: **U.S. NAVY 97-12345**



Rear:

The agency identification and military registration number (USN) shall be readily identifiable from the rear (E.g., rear or side of rear ramp/beavertail, etc.) and arranged in a single line equidistance from the vehicles outside edges. “U.S. Navy” on the left (driver side rear) followed by the USN registration number on the right (passenger side rear).

- Font Format: Stencil.
- Size & Stroke: 3-inches high, ½-inch stroke.
- Dash: ½-inch high, 1-inch long.
- Color: Matte Black.
- Example: **U.S. NAVY** **97-12345**



When hydraulic decks, tailgates, ramps, or other attachments may obscure the rear USN registration number, dual markings must be applied so that an identification number is visible regardless of the position of tailgate, ramp, or other.

When small surfaces are unsuitable or space is limited for marking sizes shown (E.g., at rear), the OEM shall review and confirm use of a shortened military registration name (E.g., "USN") with the U.S. Navy to suit the space to include location and placement:

- Example – **USN 97-12345.**

Alternative use of decals, laser engraved, riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN for all markings.

Any trailers requiring concealed markings shall be furnished with a name/data plate only, to be permanently attached to a concealed but readily accessible interior surface of the trailer, to the underside of the lid of a lockable toolbox, or in another concealed area.

WHEEL/RIM:

Wheels/Rim shall be prepped, painted, and finished based on industry standard (Matte black or flat lusterless green #34094 per Paint color specification requirements).

TIRE PRESSURE MARKING:

Tire pressure shall be stenciled/painted matte black on visible area of frame/side beams each side between the 1st and 2nd axle wheel well.

Marking shall be in following Font Format, Size, and Stroke:

- Font Format: Stencil
- Size and Stroke: 1-inch high 5/32-inch stroke.
- Dash: ½-inch high, 1-inch long.
- Color: Matte Black
- Example: **TP – 120** PSI.



Alternative use of decals, laser engraved, riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN for all markings.

LIFT AND TIE DOWN PROVISION MARKING:

LIFT POINT MARKING

Lift provision markings shall be as follows.

Lift provisions and hardware (for air, sea, and highway) shall be painted flat lusterless green #34094, in accordance with SAE AMS-STD-595A (No gloss).

Lift Provisions shall be identified/marked via stencil/painted matte black on a visible area of frame or side beam next to the lift provision.

Lift devices shall be identified by the words **LIFT HERE**.

Markings shall be in the following Font, Format, and Size:

- Font Format: Stencil.
- Size and Stroke: 1-inch high ½ inch stroke.
- Color: Matte Black.
- D-Ring Color: Flat lusterless green #34094.
- Example: LIFT **HERE**



Alternative use of decals, laser engraved, or riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN.

TIE DOWN MARKING

Tie Down provision markings shall be as follows.

Tie-down provisions and hardware (for air, sea, and highway) shall be painted flat lusterless green #34094, in accordance with SAE AMS-STD-595A (No gloss).

Tie down markings shall be identified via data plate and stencil/paint, shall clearly indicate that the points/attachments are intended for the tie down for each unit of equipment on a carrier when shipped.

Tie Down devices shall be identified by the words **TIE DOWN**.

Markings shall be in the following Font, Format, and Size:

- Font Format: Stencil
- Size and Stroke: 1-inch high by ½-inch stroke
- Color: Matte Black.
- D-Ring Color: Flat lusterless green #34094.
- Example: TIE **DOWN**



Alternative use of decals, laser engraved, or riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN.

FUEL TANK MARKING:

Fuel Storage Tank shall be marked/labeled as; Fuel Type: Diesel (JP8/F24/High Sulphur).

Alternative use of decals, laser engraved, or riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN.

HYDRAULIC TANK MARKING:

Hydraulic Storage Tank shall be marked/labeled as Oil type: MIL-PRF-2104J (OE/HDO-10) Lubricating Oil, Engine Grade 10W.

Alternative use of decals, laser engraved, or riveted, or welded plates by OEM in place of stencil/paint must be submitted for review/approval by USN.

MARKING EXAMPLES:



DATA PLATES:

All data, shipping, transportation, and Instruction plates shall be metal, non-ferrous material, and mounted on all trailers using screws, rivets, or other mechanical devices IAW NAVFAC requirements.

No sticker type data plates shall be acceptable. The data plates shall be inscribed or etched.

Safety decals must be of high quality for long lasting durability.

The following information shall be embossed /engraved onto the plates:

Nomenclature: As applicable to the type of vehicle specified in the contract (E.g., Trailer Lowbed Hydraulic RGN).

Manufacturer name, make, and model:

Manufacturer's serial number:

U.S. Navy registration number (USN):

National Stock Number (NSN):

Vehicle curb weight: _____ kg (pounds).

Payload, maximum: _____ kg (pounds).

Gross vehicle weight rating (GVWR) max. _____ kg (pounds).

Gross combination weight rating, max. _____ kg (pounds).

Engine Manufacturer name, make, and model: _____

Engine Manufacturer's serial number: _____

Date of delivery: (month and year). _

(Date of delivery shall be left blank until the vehicle is accepted by the Government inspector.

Date of delivery shall not be prior to date of acceptance indicated on form DD 250.)

Contracting Agency:

Contract number:

Warranty: _____ mo. _____ km (miles).

U.S. Property

3.8. TECHNICAL SUPPORT:

The offeror shall provide access to customer and warranty service, as well as technical support via email, phone service, internet/web, or other method.

Initial response for technical support from the offeror or certified dealer shall be within three (3) business days.

An after-hours emergency phone number shall be made available to provide first response assistance to help resolve emergency service /maintenance issues.

Parts supportability shall be provided utilizing commercial-off-the-shelf (COTS) parts to the maximum extent possible

3.9. OPERATOR, MAINTENANCE, PARTS, SERVICE MANUALS, AND SUPPORT

All printed and computer materials shall be written in English and must be able to be recreated by government for government use only and cannot be proprietary in the use with government only

Operator and Maintenance instructions manuals and original equipment manufacturer parts listings shall be provided in the following formats:

PDF reader / search capable on a CD or web-based access.

Traditional paper copies.

Operator's manuals shall contain procedures for the safe operation of all system and sub-system components along with associated safety cautions and warnings applicable to equipment operation.

Operator's manual to include 1(800) phone number, email, or other method for 24/7 access.

Service Maintenance Manual(s) shall contain:

Original equipment manufacturer's preventive maintenance schedules, procedures, and safety precautions applicable to each maintenance task.

Instructions and procedures for inspecting, testing calibration, service, repair, replacement, and overhaul of all installed components, equipment systems, and sub- systems (E.g., trailer, brakes, engine, electrical system, etc.)

Parts Manuals shall be an illustrated parts breakdown, include military modification items, and shall contain:

A pictorial representation (i.e., exploded view) of all system and sub-system components, indentured to the item level, identified by individual item numbers.

Shall list complete item names, original equipment manufacturer part numbers, quantity per end item, and the quantity per assembly.

Shall provide manufacturer name, cage code, and part numbers for all replacement parts.

Supplemental Manuals for all work tool attachments shall be included as applicable.

Operator/Maintenance/Parts manuals to include Point of Contact phone number, email, or other method to respond to customer questions/requests for manual support.

The following shall be delivered 30-45 days After Receipt of Order (ARO)

One (1) Standard Commercial off the Shelf (COTS) Product Manual, with parts list, for past build or similar equipment/product configuration to allow customer to begin Product Support Development.

The following shall be delivered 150-180 days After Receipt of Order (ARO)

One (1) set of Tech Manuals (PDF reader / search capable with **editing capabilities** on a CD or web-based access).
The contractor shall provide details of the major systems and components in the draft,
(i.e.: axles, brakes, wheels, pony engine, etc.)

Prior to submission of the DRAFT Technical Manual, individual component cut sheets may be offered or provided by OEM as they are finalized. Final as Built Tech Manuals shall include same component cut sheets if submitted.

To:

Naval Facilities Engineering Command (NAVFAC)
Engineering and Expeditionary Warfare Center (EXWC)
1000 23rd Avenue, Bldg. 1100
Port Hueneme, CA 91343
Attn: Ground Vehicles (ExPO EX022)
(POC to be provided)

The final versions of the manuals shall be completed and delivered after completion of 1st Article review and testing at final destination location but a current version although not the final needs to be onsite at the time of the 1st Article review for referencing; If any items are found at the time of the 1st Article review that need correcting or require reconfiguration of any sort that affects the manuals data will; the manuals will need to be updated and delivered first in order to receive a final acceptance.

Final As Built:

One (1) set of Tech Manuals (PDF reader / search capable with editing capabilities on a CD or web-based access)

To:

Naval Facilities Engineering Command (NAVFAC)
Engineering and Expeditionary Warfare Center (EXWC)
1000 23rd Avenue, Bldg. 1100
Port Hueneme, CA 91343
Attn: Ground Vehicles (ExPO EX022)

The following shall be delivered WITH each trailer:

One (1) set of Tech Manuals (CD or web-based access and paper copy format).

OEM Quality control print out to ship with each trailer.

OEM End of line report to be provided with manuals.

Onboard Tools (E.g., Breaker bar, Lug wrench handle, and socket tool to remove wheels.

The CD-ROMs or web-based access will incorporate a navigation page with electronic links to the Operator's Manual, Service Manual, Parts Manual, and Warranty information, as well as instructions on how to use the manual. The copy will include a table of contents with links to the specified documents or illustrations.

The OEMs shall allow for reproduction and distribution of Proprietary and/or Copyrighted Data, for government purposes and include a parts list (E.g., bolts, filters, gaskets) and a list of Collateral Items as applicable (E.g., Spare tire, spare rim, cargo straps, chains, crank handles, etc.). Parts and collateral lists shall be included as part of the Technical Manual.

Provide a Build Sheet or line setting ticket for each trailer, listing manufacturer, VIN, component part names, part numbers, serial numbers, and UID tag numbers.

3.10. GSA FORM 1398

Unless otherwise specified, the Successful offeror shall affix one copy of GSA Form 1398, GSA Purchased Trailer, after final inspection is made. Copies of GSA Form 1398 are available from the Contracting Officer.

3.11. WORKMANSHIP

Trailers shall be free from defects that may impair their serviceability or detract from appearance.

All bodies, systems, equipment, and interfaces with the chassis shall be done in accordance with the OEM's Body Builders Book.

Whenever dissimilar metals are used, they shall be insulated against corrosive action.

All components shall be new. Defective components shall not be furnished.

Component parts and units shall be manufactured to definite standard dimensions with proper fits, clearances, and uniformity.

All components shall be suitable for the intended purpose and shall be free of any characteristics or defects in material and workmanship, which may affect the performance, function, durability, and serviceability of the finished trailer, or detract from its appearance.

The purchaser reserves the right to make the final determination of the suitability of all components and their arrangement on and in the trailer.

Parts, equipment, and assemblies, which have been repaired or modified to overcome deficiencies, shall not be furnished without the approval of the purchaser.

All items shall be of good quality, in safe operating condition. All safety devices shall be fully functional; to include all manual and automatic emergency shut-off devices. All safety guards, warning labels, and safety related instruction labels shall be properly affixed at all times.

All welding, to include welded, bolted, and riveted construction utilized, shall be in accordance with the highest standards of industry and must conform to Structural Welding Code for Steel and Stainless IAW American Society of Mechanical Engineers (ASME) and American Welding Society (AWS) Standards.

All welds shall be free of excess slag and splatter.

All welds shall be uniform, have adequate penetration, and be free of voids.

All rubber components shall be ozone and UV resistant.

General appearance of the trailer shall not show any evidence of poor quality or workmanship.

The following shall be reason for rejection:

1. Rough, sharp, or unfinished edges, burrs, seams, corners, and joints.
2. Non-uniform panels. Edges that are not radiused, beveled, etc.
3. Improper preparation of steel, frame, or other parts and equipment to prevent rust or corrosion
4. Paint runs, sags, streaks, wrinkles, pin holes, craters, grit, seeds, orange peel, "fisheyes", etc. and any other imperfection, non-uniformity, improper application, or lack of complete coverage of primers, paints, or coatings.
5. Body panels or components that are uneven, unsealed, or contain cracks, dents or have voids.
6. Misalignment of body fasteners, glass, viewing panels, light housings, other items with large or uneven gaps, spacing etc. such as doors, body panels and hinged panels.
7. Improperly fabricated and routed wiring or harnesses, and electrical connections. Wiring should be adequately protected and mounted so as not to interfere with prime mover connection or operation of the trailer.
8. Improperly supported or secured hoses, wiring harnesses, mechanical controls, including interference with other components. Cables must flex between moving parts, with supporting clips securely mounted and cables secured in a permanent manner. Zip tied cables to hold ABS or other wiring in place are not acceptable.

9. Interference of chassis components, body parts, doors, etc.
10. Leaks of any gas, vacuum, or fluid lines (air conditioning, coolant, oil, etc.).
11. Noise, panel vibrations etc.
12. Inappropriate or incorrect use of hardware, fasteners, components, or methods of construction.
13. Incomplete or improper welding, riveting or bolting, not IAW ASME and AWS standards
14. Lack of uniformity and symmetry where applicable.
15. Loose, vibrating, abrading body parts, components, subassemblies, hoses, wiring harnesses or trim.
16. Improper body design or interface between prime mover and the chassis that could cause injury during normal use or maintenance, and which fail to provide access to perform routine or mandatory repairs or maintenance on trailer electrical and mechanical systems. In addition, the improper combination of options which by their combination and installation are inherently incompatible with regard to function or safety.
17. Incomplete or incorrect application of rust proofing, corrosion prevention coatings, or undercoat to all parts, equipment, and applicable surfaces.
18. Visual deformities and equipment malfunctions.
19. Unsealed appurtenances, accessories, or other body components, gaskets, etc.
20. In addition, any deviation from specification requirements or any other item, whether or not stipulated herein, that affects form, fit, function, finish, durability, reliability, safety, performance, or appearance shall be cause for rejection.

3.12. VERIFICATION REQUIREMENTS

The contractor shall host a design review meeting onsite to include GSA and the government in attendance.

The contractor shall verify compliance to the requirements herein prior to first trailer production release. Verification may be in the form of contractor production line bill of material, drawings, instructions, test results, vendor documentation, certifications, or design analytical data.

The contractor shall not proceed with production before receiving agreement from the Government that verification of compliance is satisfactory. Acceptance of designs by the Government does not relieve the contractor from the contractor's responsibility to meet all the requirements and specifications herein, whether reviewed by the Government or not.

It shall be the contractor's responsibility to assure all production documentation results in the production of trailers that are in compliance with the requirements and specifications:

PAC	Post Award Conference	Receipt of Order + 5 Days
VDR	Verification Design Review	Receipt of Order + 60 days
FDR	Final Design Review	Receipt of Order +90 Days
VI	Trailer Inspection	
BOM	Bill of Material	

The type of verification to be submitted for acceptance is listed below:

ET Engineering Test – to be performed by the contractor and results accepted by the Government. Engineering tests are tests resulting in data output from engineering instruments used to take measurements of parameters such as temperatures, sound levels. Verification may be in the form of an engineering report, test result sheet, or test summary with results documented.

COC Confirmation of Compliance – Validation by analysis or production builds documentation to be performed by the contractor and submitted for acceptance by the Government. Verification may be in the form of analysis summaries, build documentation, purchase orders, or similar written documentation.

QAPVT Quality Assurance Performance and Verification Testing – Testing to be performed and verified by the contractor to the Government. QAPVT tests are functional tests that do not involve instrumentation. The results are observable and recorded as pass/fail, i.e., trailer functions as intended; lights function as intended. Systems or components that fail shall be annotated with the corrective action taken and retested until passed, however final acceptance and 1st article operational testing will be validated by the government prior to any acceptance of 1st article.

The contractor shall, within 45 days after receipt of order, submit detailed plans for the performance of all ET tests. These plans, at a minimum, shall include all instruments to be used, their calibration certification, and the procedures to be followed.

Government acceptances of verification and designs shall constitute design freezes and enable the contractor to proceed with production activities relative to the accepted designs. The design freeze shall be effective for the duration of the production of the entire quantity of trailers being procured.

3.13. OPTIONAL REQUIREMENTS

3.13.1 GOOSENECK NATO STANAG 4009

When this option is ordered, a gooseneck with a minimum swing clearance of 90.5” shall be provided. The intent and purpose of this option is to assure compatibility with tow tractors in accordance with NATO STANAG 4009

3.13.2 INITIAL /NEW EQUIPMENT TRAINING

(Option is for trailer platform/type ordered, per training location)

The Original Equipment Manufacturer (OEM) of the trailer shall develop and provide initial/new equipment operator and maintainer training at multiple live events that incorporates and leverages existing training techniques, methods, and resources. All training materials will be in English as will be the training courses

Initial equipment operator and maintainer training shall:

Include a complete equipment system overview and operations demonstration for participants designed to include review of regular maintenance and routine repair actions to be conducted by operators/military technicians.

Provide students with the knowledge and understanding of the capabilities, functions, interfaces, limitations, and operation and external diagnostics and other tests required to support the system. Training shall include information on general safety, personal protective equipment, and preventive and corrective maintenance tasks/skills and procedures.

Be of sufficient depth to ensure that students are qualified to properly operate and maintain the trailer system in the field.

Be comprised of hands-on practical application, video, and virtual instruction; or combination of multiple equivalent methods to support learning.

or combination of multiple equivalent methods to support learning. All training materials will be in English as will be the training courses.

Take place onsite at a government facility, or other designated scheduled location via mutual agreement between the Contractor and Government (E.g.: Port Hueneme, CA; Gulfport, MS; Hampton Roads/Norfolk/Yorktown area, VA).

INITIAL training shall consist of:

Four (4) days total consisting of two, 2-day training sessions held consecutively at each location.
(to be determined via mutual agreement between the Contractor and Government-see above).

Each 2-day training session will consist of:

- One half day of training classroom instruction on Operation (knowledge) to include review of the Operations, Maintenance, and technical manual and,
- One and a half training days of practical field/hands on application (skill) to include instruction on maintenance/repair of trailers.
- Each training day shall include two 15-minute breaks and one, 1-hour lunch break.

The Contractor shall provide operator and maintainer training for up to fifteen (18) formal students, for each 2-day training session, to include Military/Government instructors and key personnel.

The OEM shall provide the Government two electronic (MS Word) copies of any training material including references, intended to be utilized or required to conduct the course(s)/event instruction/training, at least forty-five (45) days prior to training to the Government. This shall include lesson plans, student guides, videos, Visual Information (MS Power Point) presentations, course tests, or other.

Operator/Technical Manuals may be utilized as alternative training materials.

All developed training materials shall be delivered to the Government (NAVFAC EXWC/TSA) in electronic format by electronic means or on CD-ROM media.

The Government shall be granted right to reproduce training material.

The Government will provide comments/edits within ten (10) business days for contractor adjudications.

The OEM will resubmit a "Final" version of all training material to the Government (as a package) within 10 working days prior to the first initial training event.

As part of the training package, the OEM shall provide hard copies of Student Guides (Operator and Maintainer) and all references and training material required to conduct the training at the time of the respective training events. There shall be enough hard-copy Student Guides for key personnel (to include instructor and observers).

The OEM shall provide all materials required for the training of Government personnel, to include actual materials, parts (E.g., bolts, hoses, tools) to be used during the training for contractor technician to demonstrate removal and replacement of parts, as applicable.

The OEM shall not put time constraints or expiration dates on purchased training. Specific training dates and daily training times shall be coordinated between the Government (NAVFAC EXWC/TSA) and the OEM.

3.13.3 ADDITIONAL Operator/Maintainer Equipment Training **(Optional: For trailer platform/type ordered, per training location)**

The Original Equipment Manufacturer (OEM) shall provide Additional Equipment operator and maintainer training at multiple live events that incorporate and leverage existing training techniques, methods, and resources.

ADDITIONAL Operator/Maintainer Equipment Training shall:

Include a complete equipment system overview and operations demonstration for participants designed to include review of regular maintenance and routine repair actions to be conducted by operators/military technicians.

Provide students with the knowledge and understanding of the capabilities, functions, interfaces, limitations, and operation and external diagnostics and other tests required to support the system. Training shall include information on general safety, personal protective equipment, and preventive and corrective maintenance tasks/skills and procedures.

Be of sufficient depth to ensure that students are qualified to properly operate and maintain the trailer system in the field.

Be comprised of hands on, practical application, video, and/or virtual instruction; or combination of multiple equivalent methods to support learning.

Take place onsite at a Government Facility, or other designated scheduled location via mutual agreement between the Contractor and Government (E.g.: Port Hueneme, CA; Gulfport, MS; Hampton Roads/Norfolk/Yorktown area, VA).

ADDITIONAL Operator/maintainer Equipment Training shall consist of:

Four (4) days total consisting of two, 2-day training sessions held consecutively at each location.
(to be determined via mutual agreement between the Contractor and Government-see above).

Each 2-day training session will consist of:

- One half day of training classroom instruction on Operation (knowledge) to include review of the operations, maintenance, and technical manual and,
- One and a half training days of practical field/hands on application (skill) to include instruction on maintenance/repair of trailers.
- Each training day shall include two 15-minute breaks and one, 1-hour lunch break.

The Contractor shall provide ADDITIONAL operator and maintainer training for up to fifteen (18) formal students, for each 2-day training session, to include Military/Government instructors and key personnel (should option be initiated).

The OEM shall provide the Government with two electronic (MS Word) copies of any training material, including references, and any updates made after initial operator/maintainer training, intended to be utilized or required to conduct the course(s)/event instruction/training, at least **forty-five** (45) days prior to training to the Government. This shall include lesson plans, student guides, videos, Visual Information (MS Power Point) presentations, course tests, or other. Operator/Technical Manuals may be utilized as alternative training materials.

All developed training materials shall be delivered to the Government (NAVFAC EXWC/TSA) in electronic format by electronic means or on CD-ROM media.

The Government shall be granted right to reproduce training material.

The Government will provide comments/edits within ten (10) business days of notification for contractor adjudications.

The OEM will resubmit a “Final” version of all training material to the Government (as a package) within 10 working days prior to the first additional training event.

As part of the training package, the OEM shall provide hard copies of Student Guides (Operator and Maintainer) and all references and training material required to conduct the training, at the time of the respective training events. There shall be enough hard-copy Student Guides for key personnel (to include Instructor and observers). All training materials will be in English as will be the training courses.

The OEM shall provide all materials required for the training of Government personnel, to include actual materials, parts (E.g., bolts, hoses, tools, etc.) used during the training for contractor technician to demonstrate removal and replacement of parts, as applicable. All training materials will be in English as will be the training courses.

The OEM shall not put time constraints or expiration dates on purchased training. Specific training dates and daily training times shall be coordinated between the Government (NAVFAC EXWC/TSA) and the OEM.

4.0. TESTING, INSPECTION, ACCEPTANCE

4.1 TRAILER WEIGHT

The completed trailer shall be weighed on a certified scale, to determine the curb weight and axle weights.

4.2 PRE-SERVICE INSPECTION

The successful offeror shall perform a pre-service inspection of the production trailers based, as a minimum, on TMC RP 803- Pre-Service Trailer Inspection. All items listed in RP 803 shall be inspected and verified as satisfactory with all adjustments noted and verifications that all needed repairs have been made. The successful offeror's own form may be used in lieu of the form in RP 803 if the successful offeror's inspection form includes, as a minimum, all the inspection criteria in RP 803 applicable to the trailer. A copy of this completed inspection form shall be made available to any government inspector and shall be kept in the trailer as part of the documentation shipped with the trailer.

4.3 VOCATIONAL TEST

All equipment shall be functionally tested and verified to operate in accordance with manufacturer's operating instructions and with all government prime movers listed above at time of operational testing.

4.4 OPERATIONAL TESTING AT GULFPORT, MS

The First Production Vehicle shall be shipped to the Navy's identified location. The Navy will conduct operational testing to verify compliance with the operational and performance requirements referenced within this Purchase Description with all listed Navy Prime movers which are listed in section 3.0 here in.

4.4.1. The operational testing conducted by the Navy shall include:

Verification of form, fit and function with no obstructions or interface operability issues with the specified Prime movers in section 3.0. 1st article operational testing will have the OEM send the trailer to testing site and provided it passes testing it will be accepted. If testing is not passed it will be the responsibility of the contracted agent to send trailer to facility to get items corrected at their cost and then sent back for operational testing until the trailer passes 1st article operational testing. After the 1st trailer is accepted and passes operational testing, all follow on trailers will be required to meet the same standard as the 1st article trailer at time of delivery to the end user and if not the contracted agent is responsible for correcting to get to that standard at the contracting agents cost to include all shipping back and forth until shipped and accepted by end user. Note final acceptance will not be given until ATTLA certification is provided by the contracted agent.

4.4.2. Verification to transport a combination of various sized ISO 668 shipping containers and/or containerized freight, breakbulk cargo, construction equipment (E.g. Skid steer loaders, Generators, Light plants, HMMWV's), equipment attachments, palletized supplies, and/or

miscellaneous construction supplies, configured in a single load, distributed evenly across the load space, as specified in section 3.1. Additionally, testing will be conducted with all prime movers to test wet line kit interface, electrical and prime mover heights and swing clearance along with braking system hookups.

4.4.3 Verification of performance as referenced in section 3.1, 4.4.1. and 4.4.2. including:

4.4.3.1 Operation on cross country on highways, paved secondary, gravel, unimproved, or cobblestone roadways

Ability to traverse on, through, over, and across multiple on-and-off road terrains, surfaces, roads, streets, parking aprons, runways, airfield shoulders and overruns, parking and open storage areas, for transfer transport, or on/offloading of cargo, equipment, or material.

4.4.3.2 Capability of operating in a wide variety of climatic conditions and environments, able to function in operating temperatures ranging from -25 degrees to 130 degrees Fahrenheit.

4.4.3.3 Ground Clearance: Should be sufficient to prevent damage to the axles or frame or any other trailer surface when trailer is in operation on unimproved terrain while loaded.

Capability of loading to dock height of 60 Inches (not for outsized equipment)

Verification of final weights to complete the ATTLA certification process and due before trailer delivery to end user

The contractor and their suppliers are welcome to attend and witness the operational testing activities conducted by the Navy

The estimated time to complete the operational testing is 5 business days or until testing is complete due to any delays dependent on weather and unforeseen circumstances i.e. Prime mover breakdown

The Navy shall notify the Contracting Officer of the results of their operational testing.

Pending successful results, the Contracting Officer will provide notification of the Navy 's final acceptance of the First Production Vehicle. The Contracting Officer will provide notification to proceed with production of the balance of orders.

Pending unsuccessful results, the Contractor and their supplier(s) shall remedy any non-conformances prior to government's acceptance. Retesting may be required.

5.0 WARRANTY

The successful offeror shall provide a 60-month comprehensive bumper to bumper warranty.

The warranty shall include the trailer manufacturer's commercial warranty, including the body and all other parts and components required herein, against parts failure or malfunction due to design, construction or installation errors, defective workmanship, and missing or incorrect parts, for a minimum period of 60 months from date of delivery of destination.

If the successful offeror receives from any supplier or subcontractor additional warranty on the whole or any component of the trailer, in form of time, including any prorated arrangements, or the successful offeror generally extends to its commercial customers greater or extended warranty coverage, the government shall receive corresponding warranty benefits.

The warranty coverage shall begin when the Government accepts the trailer from the successful offeror at destination.

The offeror shall provide access to warranty service via email, phone service, internet/web, or other method.
Initial response for on-site warranty repairs from the offeror or certified dealer shall be within three business days.

Cost for parts and shipping for warranted repairs shall be the responsibility of the selected vendor. This shall include 3rd party Original Equipment Manufacturer (OEM) selected vendors for corrective action or warranty issues on the trailer if applicable.

If service or repair is needed, the offeror will provide an authorized location within the Continental United States (CONUS), to allow for staged equipment to be repaired and will supply parts and labor for all items under standard warranty agreement.

OEM replacement parts shall be shipped via the most expedient shipping method (E.g., FedEx, DHL, UPS, etc.) to CONUS Government locations for distribution.

The successful offeror shall furnish the Government with warranty certificates showing evidence that the warranty requirements above have been met.

Warranty registration card shall also be provided as applicable.

6.0 POST AWARD MEETINGS

The successful offeror shall host and participate in a post award review meeting (PAC/VDR/FDR). Attendees typically include, but are not limited to, GSA engineering personnel, Navy and the end user representative(s), and GSA contracting personnel. The meetings will be scheduled by the GSA as soon after award of contract as practicable.

6.1 DELIVERY LOCATIONS

The trailers shall be delivered (FOB Destination) to the locations identified on the applicable GSA Motor Trailer Delivery Order.

All drivers shall possess a valid form of federal identification (license, TWIC card, Veterans ID card). If the drivers' Commercial Driver License (CDL) states "NOT FOR FEDERAL IDENTIFICATION", the driver will not be allowed on Federal Property. The drivers shall show proof of insurance.

The contractor may stack trailers (wheels down only) for transport to destination. The contractor shall be responsible for offloading and presenting the trailers for final destination inspection with wheels on ground. The Navy shall inspect for damage in transit prior to acceptance at destination and shall submit claims in accordance with the offeror's proposed damage in transit terms and conditions.

The contractor shall contact the NAVFAC EXWC point of contact no later than 30 days prior to delivery to validate DELIVERY ADDRESS information. Receiving officers will coordinate physical equipment deliveries with transportation drivers.

**NSWC W8Y0AA-(3 EA)
NSW OPSTOCKS
106 Jacobson Drive
Poca, WV 25159
Mark For: NSW-OPSTOCK01
POC: Paul D Parsons
Comm:304-561-6885**

**NAVELSG N81464- (10 EA)
Naval Expeditionary Logistics
Support Group
593 Mayfield Street Cheatham Annex
Williamsburg, VA 21385-5831
POC: Jonathan Pruett Cell: 757-343 1174**

NBG N69433 (2 EA)
USMC Blount Island Command
6200 Flagship Circle
Jacksonville, FL 32226-3404
Mark For: TA5501

POC: Leslie Lavarez
Comm: 904-696-3783
POC: Clarence Hendrickson
Comm: 904-696-5124

Point of Contact is:
NAVFAC Engineering and Expeditionary Warfare Center (NFEXWC)
1000 23rd Ave, Bldg. 1100
Port Hueneme, CA 93043
Attn: Ground Vehicles (ExPO EX022)
(POC to be provided).

See Separate Shipping Instructions for more information.

6.2 NAVY HOURS OF OPERATIONS

Delivery hours are Monday through Thursday 0800 AM to 1430 PM local time.

No Friday or after-hours availability. Closed Federal holidays.

The contractor shall schedule all delivery activities with the delivery points of contact.